BCSE429P

Cyber Physical Systems Design Lab

Laboratory 7 - SUMO and Vehicular Networks

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Question:

Follow the above video for full details: (the above video was made for Ubuntu, but you can try installing it for Windows) . Also this video is for custom made roads, but you can use OpenStreetmaps, where you can select the maps across the world and emulate the vehicles on road (Step 2)

Step 1: Download SUMO Software for Windows 11.

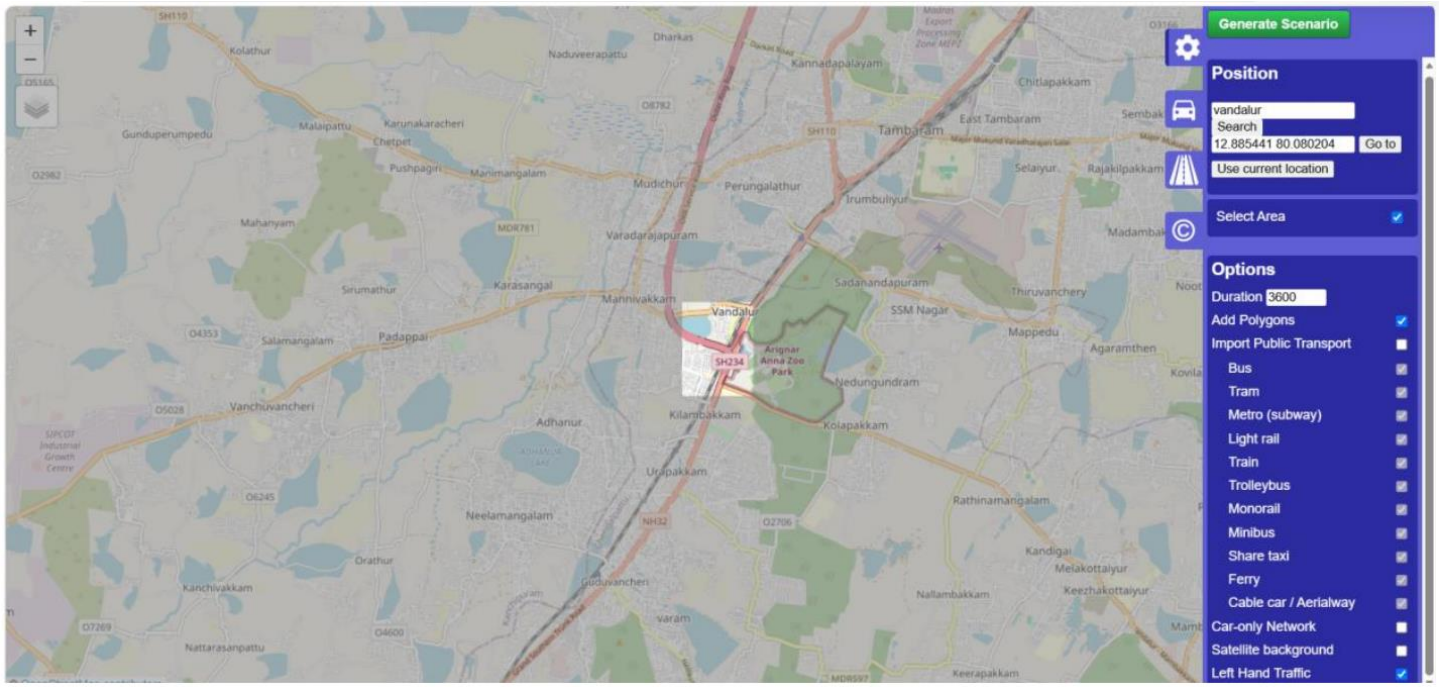
Eclipse SUMO - Simulation of Urban MObility

Step 2: Open a tool called ***osmWebWizard.py***, which opens a browser, and select an area of your choice (I tried, Guindy Chennai).

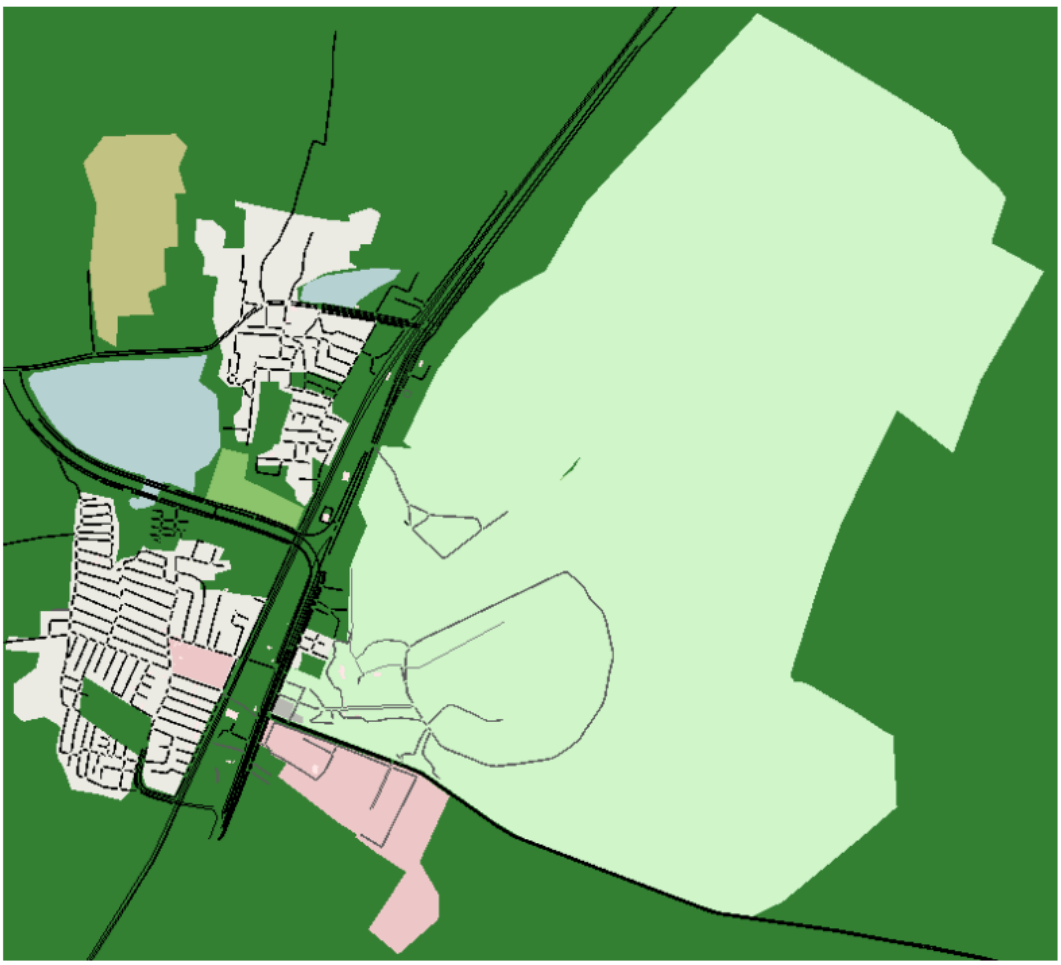
Step 3: Enable Vehicular Emission as per EURO norms to the vehicles and the software will generate the Emission data in a XML file

Step 4: the XML file to be converted to csv file using a tool called xml2csv.py which is available in SUMO software itself.

Using osmWebWizard for area selection:



Generated SUMO GUI:



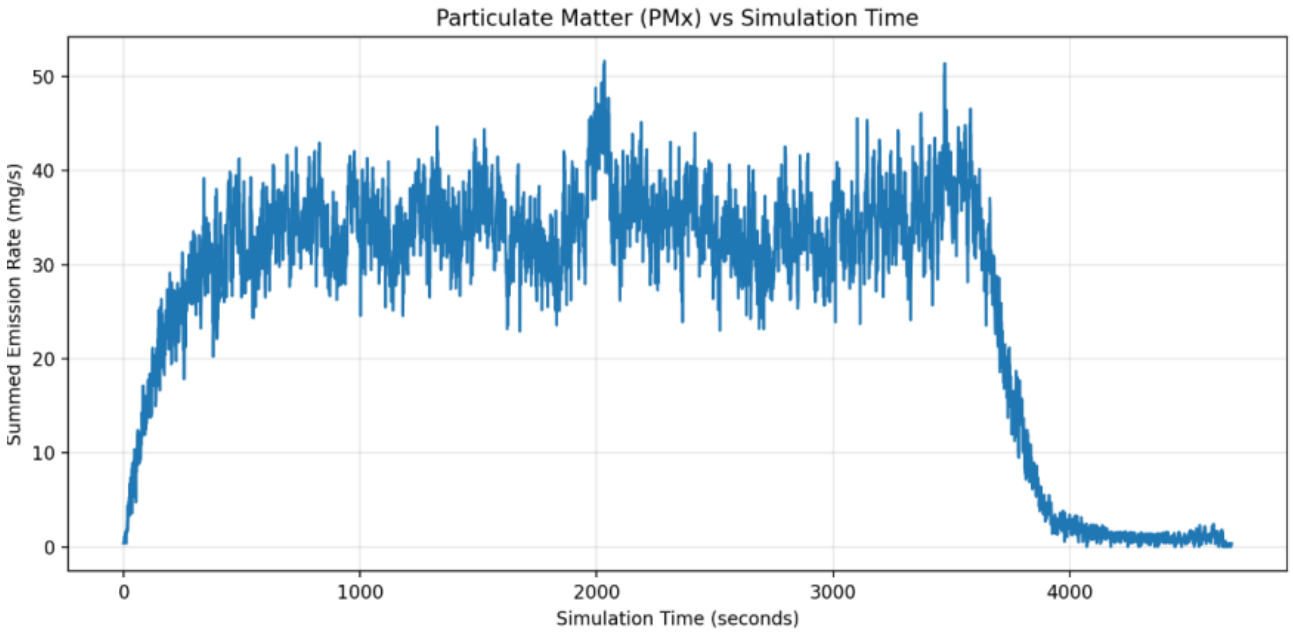
Command to generate emission:

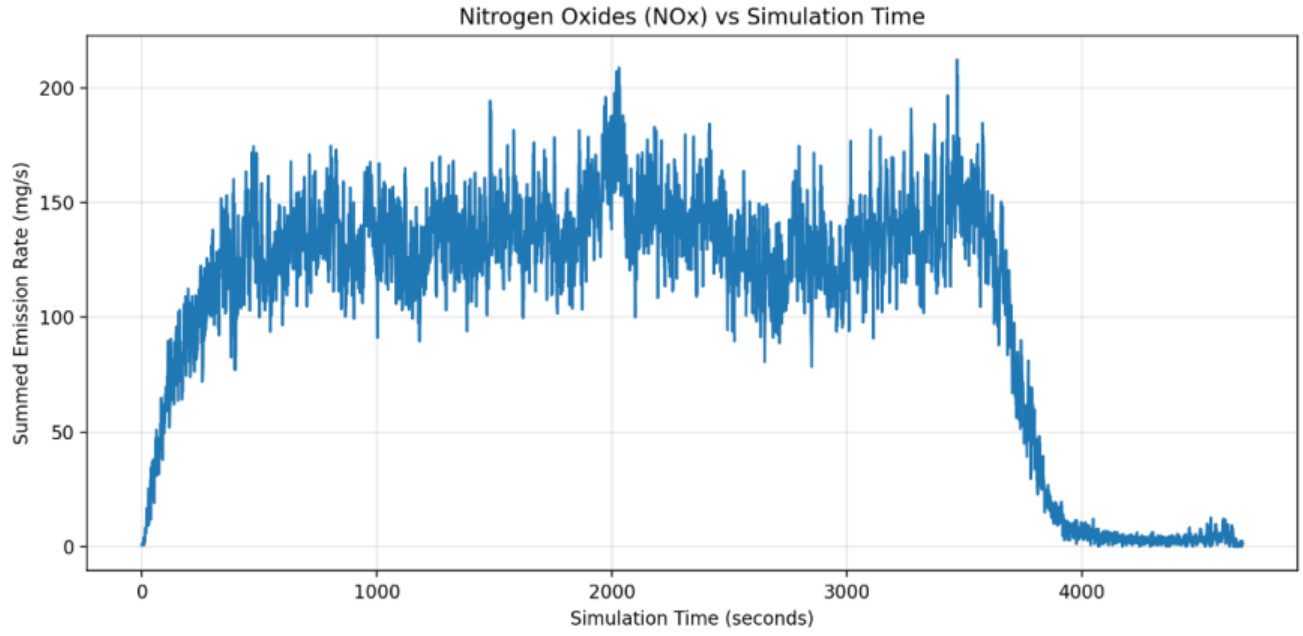
"%SUMO_HOME%\bin\sumo.exe" -c "C:\Users\Aman\2026-10-14-03-03-26\osm.sumocfg" -- emission-output "C:\Users\Aman\2026-10-14-03-03- 26\emissions.xml"

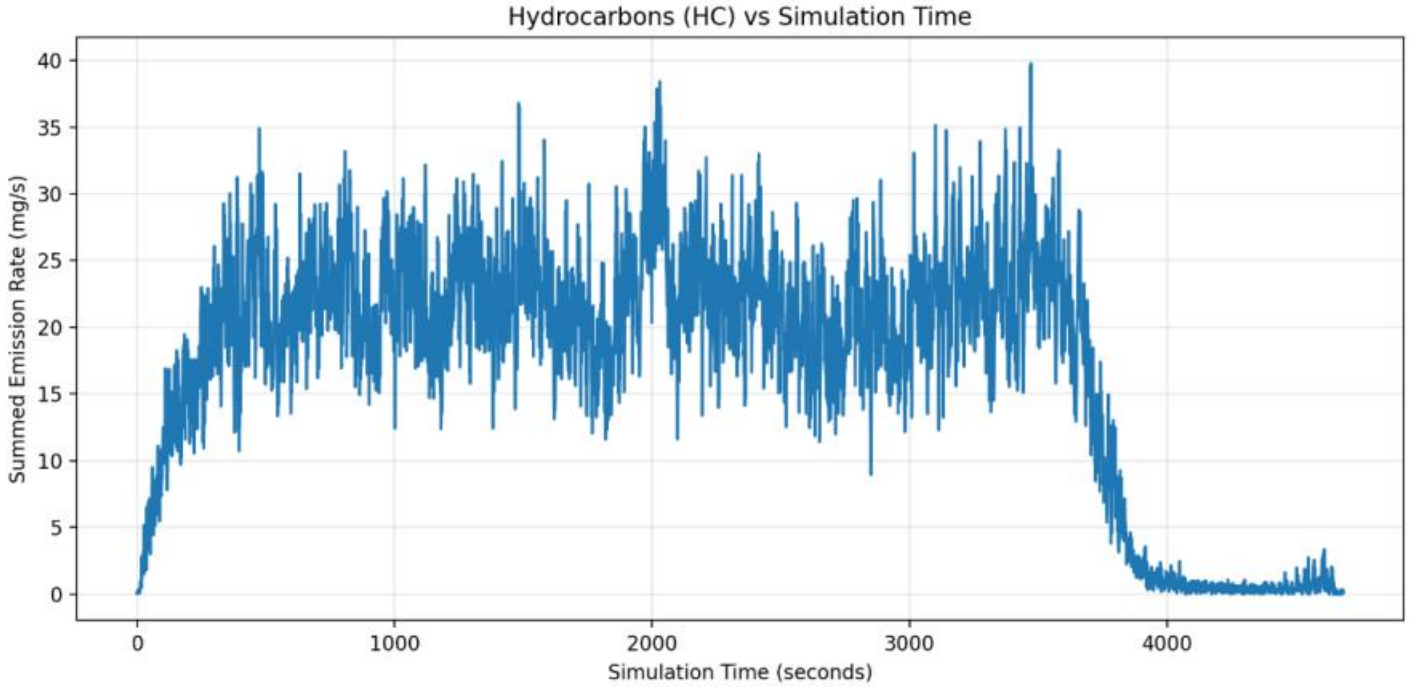
Command to convert to CSV file:

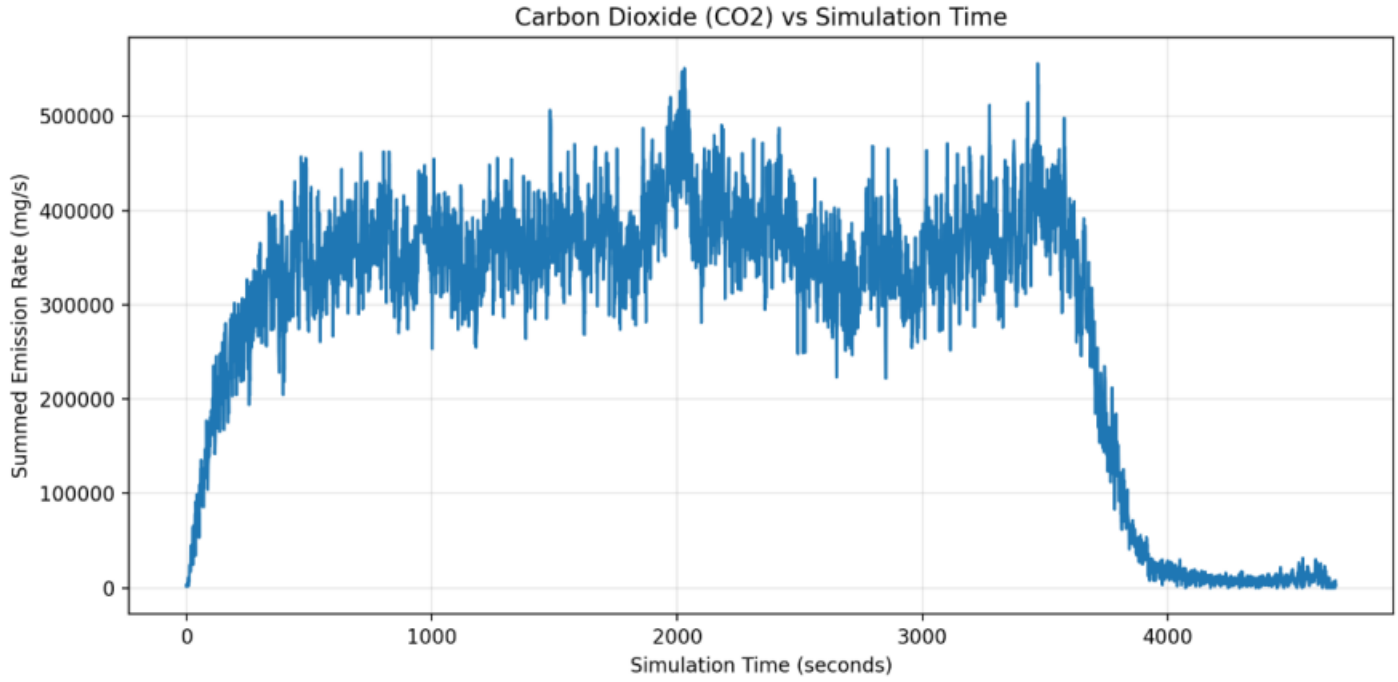
python "%SUMO_HOME%\tools\xml\xml2csv.py" "C:\Users\Aman\2026-10-14-03-03-26\emissions.xml"

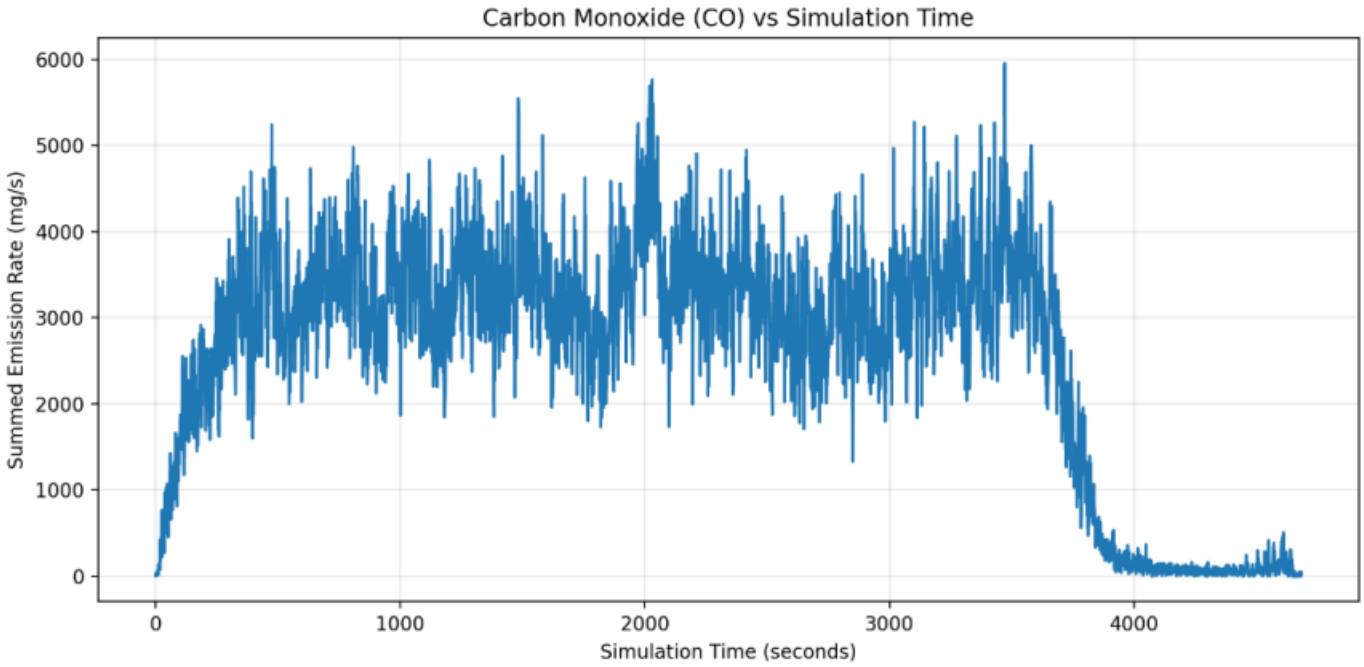
Obtained Plots:






Hydrocarbons (HC) vs Simulation Time
Summed Emission Rate (mg/s)
0
5
10
15
20
25
30
35
40
0
1000
2000
3000
4000
Simulation Time (seconds)


Carbon Dioxide (CO2) vs Simulation Time
Summed Emission Rate (mg/s)
0
100000
200000
300000
400000
500000
0
1000
2000
3000
4000
Simulation Time (seconds)


Carbon Monoxide (CO) vs Simulation Time
Summed Emission Rate (mg/s)
6000
5000
4000
3000
2000
1000
0
0
1000
2000
3000
4000
Simulation Time (seconds)